

Department of Mathematics and Applied Mathematics
3MS Metric Spaces

Tutorial 7

2024

1. In the definition of a homeomorphism, we insisted that the function had to be bijective and continuous and have a continuous inverse. Here is (another) example of a continuous bijection that does not have a continuous inverse.

Let $A = [0, 1] \cup (2, 3]$ and $B = [0, 2]$ be metric subspaces of the reals with the usual metric. Define $f : A \rightarrow B$ by

$$f(x) = \begin{cases} x & \text{if } x \in [0, 1] \\ x - 1 & \text{if } x \in (2, 3] \end{cases}$$

- (a) Draw the graph of f .
 - (b) Check that f is one-to-one and onto.
 - (c) It is clear that f is continuous at each point of $[0, 1) \cup (2, 3]$. Use the definition of continuity (using sequence convergence) to check that f is continuous at 1. (Remember that the domain of f is A , not $\mathbb{R}$.)
 - (d) Let $g = f^{-1}$. Write down a formula for g . If you feel like it, draw the graph of g .
 - (e) Show that g is not continuous at 1.
2. Let (X, d) be a metric space and $a \in X$. Define $f : (X, d) \rightarrow (\mathbb{R}, d_E)$ by

$$f(x) = d(x, a)$$

where d_E is the usual metric on the reals. Show that f is uniformly continuous.

(The characterization of uniform continuity using sequences is easy to use here. Set up what you need to assume and prove; then see p68 if you need a hint.)

3. A uniform isomorphism between metric spaces is a bijection that is uniformly continuous, and whose inverse is also uniformly continuous. In this example, you will see an example of a function which is a homeomorphism (that is, a topological isomorphism), but not a uniform isomorphism.

Let δ be the metric on $\mathbb{N}$ defined by $\delta(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right|$.

(You may take as a given (and check later) that this is a metric on $X = \{x \in \mathbb{R} : x > 0\}$ and so also a metric on $\mathbb{N}$.)

- (a) Show that δ and d_E (the Euclidean metric) both induce the discrete topology on $\mathbb{N}$. Deduce that the function $f : (\mathbb{N}, \delta) \rightarrow (\mathbb{N}, d_E)$ given by $f(x) = x$, for all $x \in \mathbb{N}$, is a homeomorphism.
- (b) Show that the function $f : (\mathbb{N}, \delta) \rightarrow (\mathbb{N}, d_E)$ is not a uniform isomorphism (defined in (a)), because it is not uniformly continuous. (Hint: Show that $(x_n) = (2n - 1)$ and $(y_n) = (2n)$ are neighbours in $(\mathbb{N}, \delta)$, but not in $(\mathbb{N}, d_E)$).

4. Consider the statement: “Prove or disprove: If (X, d) and (X, d') are isometric metric spaces, then d and d' are equivalent metrics.”

Hazard a guess as to the correct answer and then work through the following.

You know by now that, if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a one-to-one function, then d_f given by $d_f(x, y) = |f(x) - f(y)|$ is a metric on $\mathbb{R}$. (See Q3 on p27 of the notes.)

- (a) Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} x & \text{if } x < 1 \\ 2 & \text{if } x = 1 \\ x + 2 & \text{if } x > 1 \end{cases}$$

Describe the open ball $U = B(1, \frac{1}{2})$ in $(\mathbb{R}, d_g)$.

- (b) Define $h : \mathbb{R} \rightarrow \mathbb{R}$ by $h(x) = g(-x)$. Show that the map $k : (\mathbb{R}, d_g) \rightarrow (\mathbb{R}, d_h)$ given by $k(x) = -x$ is a bijective isometry.
- (c) Is the set U (from (a) above) open in $(\mathbb{R}, d_h)$? Are the metrics d_g and d_h equivalent?
- (d) Now consider the statement:
“If (X, d) and (X, d') are isometric metric spaces, then (X, d) and (X, d') are homeomorphic.” Is this true or false?

5. Let d be the usual metric on $\mathbb{R}^n$, let A be a subset of $\mathbb{R}^n$ and let d_A be d restricted to A .

- (a) Let $n = 1$. If you want d_A to be the discrete metric, how many elements can you have in your set A ?
- (b) Answer the same question for $n = 2$ and $n = 3$.
- (c) Conclude that no two of $\mathbb{R}$, $\mathbb{R}^2$ and $\mathbb{R}^3$ are isometric (i.e. metrically isomorphic).

If you felt like it, you could extend this argument to show that $\mathbb{R}^k$ and $\mathbb{R}^m$ are not isometric, for any positive integers $k \neq m$. (But you don't need to.) The fact that $\mathbb{R}^k$ and $\mathbb{R}^m$ are not homeomorphic for $k \neq m$ is much harder to prove, and we certainly won't get to it in this course.